

# SAISUMANTH TALLAPRAGADA

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## PROFESSIONAL SUMMARY

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Software Engineer with 4+ years building backend services, distributed systems, and automation pipelines using Java Spring Boot, Python FastAPI, Kafka, and cloud-native infrastructure. Owned CI/CD pipelines, production observability, and end-to-end delivery for high-volume systems, with applied AI work spanning agentic LLM pipelines, text-to-SQL agents, and ML systems.

## PROFESSIONAL EXPERIENCE

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### Cypress Creek Energy

Software Engineer Intern

June 2025 - Present

Durham, NC, USA

- Architected AWS-hosted FastAPI services ingesting enterprise IT asset data from Microsoft Graph API into Snowflake, owning end-to-end delivery through production rollout and reducing inventory discrepancies by 90% across IT operations.
- Engineered Python ETL pipelines transforming energy settlement billing data into Snowflake tables, replacing manual Excel workflows and reducing Finance team overhead by 60% through production data automation.
- Designed a multi-source workforce data pipeline integrating Sitetracker, NetSuite, ADP, and Microsoft Entra using cron-driven Python ETL and Snowflake schemas, cutting manual HR processing workflows by 70%.
- Owned observability and deployment workflows for IT asset, workforce, and revenue services using Grafana, Prometheus, Loki, Argo CD, and Azure DevOps CI/CD, cutting pipeline incident detection time from 30–45 minutes to near-instant.

### Tata Group (TCS)

Software Engineer

May 2021 - July 2024

India

- Led a team of 3 engineers building a 7-country order management backend using Java Spring Boot, REST APIs, and Apache Kafka, supporting high-volume order, payment, and receipt workflows across global enterprise operations.
- Modernized high-traffic payment services from Apache Struts to Spring Boot, implementing load balancing and high availability patterns at enterprise scale, reducing API response times from 200ms to 50ms.
- Optimized Oracle SQL and PL/SQL queries through execution plan analysis, indexing, and query tuning, reducing data retrieval latency by 30% and automating bulk product onboarding workflows that cut manual effort by 80%.
- Enforced code quality across order and payment services using JUnit, Mockito, and Swagger, improving maintainability and supporting reliable releases across a distributed cross-country engineering team.
- Drove Agile SDLC delivery using Jenkins and TortoiseSVN for CI/CD, co-owning on-call rotation for production services, triaging incidents, and sustaining release velocity across active global deployments.

### National Informatics Center

Software Engineer Intern

March 2020 - April 2020

India

- Developed a full-stack Fleet Management System using Java, Spring Boot, Hibernate JPA, PostgreSQL, jQuery, and Angular to optimize fuel expenditure tracking and maintenance workflows, managing builds with Maven and version control via GitHub.

## PROJECTS

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### LLM-Based Code Quality Analysis: Expertiza (Under Review, IEEE FIE 2026)

- Engineered an agentic LLM pipeline combining prompt-engineered reasoning chains with AST-based static analysis to detect SOLID, coupling, and cohesion violations across 722 pull requests on the Expertiza OSS platform, eliminating manual review overhead at scale.

### C++ Systems Project: Custom 2D Game Engine

- Built a modular C++17 game engine with SDL3, ZeroMQ, and CMake, implementing systems-level components including rendering, input, networking, multithreading, and custom memory management for reusable, high-performance gameplay infrastructure.

### LLM-Based AI Agent: Retail Text-to-SQL Agent

- Built an LLM-powered AI agent using Python, AutoGen, OpenAI API, SQLite, and Streamlit that converts plain-English analytics questions into executable SQL, validates queries through tool-based database execution, and self-corrects using feedback.

## SKILLS

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**Languages:** Java, Python, C/C++, JavaScript, TypeScript, SQL

**Frameworks & Libraries:** Spring Boot, JSP, FastAPI, Apache Kafka, Struts, Node.js, React.js, AngularJS, Hibernate JPA, OpenAPI/Swagger, HTML/CSS, Bootstrap, LangChain, Pytorch, JUnit, Mockito

**Databases & Warehousing:** PostgreSQL, MySQL, Snowflake, MongoDB, Oracle BI Publisher, Sigma BI

**Cloud Infrastructure & Tools:** AWS, Azure DevOps, Docker, Kubernetes, Argo CD, Jenkins, Git, TortoiseSVN, GitLab, Linux/Unix, Grafana, Loki, Prometheus, JIRA, Figma

## EDUCATION

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North Carolina State University - Raleigh

Master's, Computer Science

August 2024 - May 2026

GPA: 4.0

- Algorithms & Data Structures, Object-Oriented Design, Computer Networks, Machine Learning, AI & Data Science, Database Systems